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To cite this article: Mahmoud Abdi Tabari, Ali Kushki & Yizhou Wang (18 Sep 2025): Comparing the effects of teacher- and AI-mediated corrective feedback on accuracy, complexity, and quality in L2 written narratives, Computer Assisted Language Learning, DOI: [10.1080/09588221.2025.2561608](https://doi.org/10.1080/09588221.2025.2561608)

To link to this article: <https://doi.org/10.1080/09588221.2025.2561608>

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Comparing the effects of teacher- and AI-mediated corrective feedback on accuracy, complexity, and quality in L2 written narratives

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ABSTRACT

The rise of generative artificial intelligence (AI) has introduced new possibilities for the implementation of written corrective feedback (WCF), a cornerstone of second language (L2) writing instruction. This study investigates the comparative effects of teacher- and AI-mediated WCF on the complexity, accuracy, and quality of L2 narrative writing. Participants were 166 upper-intermediate ESL undergraduates enrolled in 12 international writing classes at a U.S. university. Of these, 80 received teacher feedback and 86 received AI feedback. All participants completed an initial narrative writing task, revised their texts based on the feedback provided, and submitted final drafts. Analyses revealed that both groups improved significantly in accuracy, with teacher feedback resulting in greater error reduction. Lexical complexity also increased in both groups, with teacher feedback prompting more lexical variation. In contrast, syntactic complexity declined across both groups, marked by shorter T-units and reduced subordination, coordination, and nominalization, but the decline was less pronounced in the AI-feedback group, suggesting a potential advantage of AI in maintaining syntactic complexity. Writing quality improved in both groups, particularly in writing conventions and overall scores, with no significant differences in other dimensions. These findings highlight AI-mediated WCF as a promising complement to teacher feedback, with distinct linguistic outcomes.

ARTICLE HISTORY

Received 5 May 2025
Accepted 11 September 2025

KEYWORDS

Written corrective feedback; ChatGPT; L2 writing; linguistic complexity and accuracy; artificial intelligence

Introduction

Timely, adaptive, and comprehensive feedback has long been recognized as a cornerstone of second language (L2) writing instruction and research, yet it remains a contested and multifaceted issue (Qin & Stapleton, 2023). Despite extensive theoretical discourse and empirical inquiry, optimizing feedback for pedagogical effectiveness

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 Supplemental data for this article is available online at <https://doi.org/10.1080/09588221.2025.2561608>

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continues to present significant challenges. Instructors often contend with practical constraints, such as large class sizes, limited time, and inflexible curricula, that require challenging trade-offs. These trade-offs involve not only deciding which aspects of student writing to prioritize but also determining the most effective ways to deliver feedback that promotes measurable improvements in writing accuracy, complexity, and overall quality.

To address these challenges, technology has increasingly been leveraged to enhance feedback efficiency in L2 writing instruction. While early automated writing evaluation (AWE) systems primarily focused on surface-level error detection (Zhang, 2020), recent advancements in generative artificial intelligence (AI) have introduced more sophisticated, interactive tools, such as ChatGPT and Copilot. These AI-driven chatbots employ machine learning to analyze linguistic patterns and provide real-time, context-sensitive feedback, offering a potential paradigm shift in instructional support (Barrot, 2023; Cardona et al., 2023). As AI-mediated feedback gains traction, its pedagogical implications warrant systematic investigation beyond speculation.

Recent empirical studies have begun to compare AI-generated feedback with traditional teacher feedback, exploring distinctions in content, effectiveness, and learner engagement (Çöp, 2025). While these comparative analyses are crucial, an equally pressing question is the extent to which AI-mediated corrective feedback affects L2 learners' writing performance, as compared to teacher-mediated corrective feedback. Addressing this gap, the present study examines the impact of teacher- and AI-mediated feedback on the accuracy, complexity, and overall quality of L2 written narratives.

Literature review

Written corrective feedback in L2 writing research

Written corrective feedback (WCF) has become a central focus in the expanding and dynamic field of L2 writing research (Hirvela & Belcher, 2024). Over recent decades, writing pedagogies have undergone significant evolution, yielding increasingly diverse and effective strategies for enhancing writing proficiency across varied learner profiles. Among these developments, the role of WCF stands out as a particularly enduring and influential area of investigation, recognized not only for its potential to improve L2 writing performance but also for its broader contributions to second language development (Bitchener & Storch, 2016).

Extensive research has examined the relative effectiveness of different types of WCF (e.g. Bitchener & Knoch, 2010) and the varied foci of feedback (e.g. Ellis et al., 2008; Sheen et al., 2009; Van Beuningen et al., 2012), approaching these from multiple theoretical perspectives. This body of work has advanced a more nuanced understanding of how WCF impacts both writing accuracy and overall language acquisition.

Particularly, the scope of WCF research has gradually expanded beyond a primary concern with grammatical accuracy to encompass a wider range of error types and learner needs. Decades of empirical inquiry into prompting strategies, especially the comparison between direct and indirect feedback, have generated insightful yet

sometimes conflicting findings. Some scholars advocate for indirect feedback, emphasizing its potential to promote learner autonomy by encouraging self-identification and correction of errors (Ferris, 1995; Ferris & Roberts, 2001). In contrast, others highlight the efficacy of direct feedback, particularly when addressing more complex or persistent errors (Bitchener & Knoch, 2010).

Due to variability in study designs, learner populations, and instructional contexts, the literature has yet to yield definitive conclusions favoring one feedback approach over another (Bitchener, 2012; Kang & Han, 2015), underscoring the complexity of optimizing WCF in diverse L2 writing settings.

Advances and challenges in written corrective feedback: human and AI approaches in L2 writing

Research on WCF and its effects on L2 writing has increasingly adopted a comprehensive perspective, exploring multiple nuanced dimensions of learners' written production. A well-established consensus identifies L2 writing proficiency as consisting of four interrelated subsystems: syntactic complexity, accuracy, lexical complexity, and fluency—collectively known as CALF (Abdi Tabari, 2021; Housen et al., 2012). Accuracy refers to 'the extent to which an L2 learner's performance and the L2 system that underlies this performance deviates from a norm' (Housen et al., 2012, p. 4). Syntactic and lexical complexity denote 'the capacity to use more advanced language, with the possibility that such language may not be controlled so effectively' (Skehan & Foster, 1999, p. 96). Fluency captures 'the capacity to use language in real-time, to emphasize meaning, possibly drawing on more lexicalized systems' (Skehan & Foster, 1999, pp. 96–97). As pointed out by Abdi Tabari et al. (2024), these four dimensions represent key processes in L2 development: acquiring new linguistic elements (complexity), refining and restructuring knowledge (accuracy), and consolidating proceduralized language use (fluency).

Extensive research has examined the influence of WCF on these linguistic facets beyond accuracy, yielding mixed findings (see [Supplementary Appendix A](#)). Some studies report significant gains in syntactic and lexical complexity following revisions prompted by WCF (Robb et al., 1986; Sheppard, 1992), while others observe minimal or no changes (Hartshorn & Evans, 2015; Van Beuningen et al., 2012). These discrepancies may stem from differences in the nature of feedback provided: indirect WCF is suggested to foster deeper cognitive engagement, whereas direct WCF may better address complex errors. Similarly, regarding fluency—often measured by words per minute—some research documents improvements after WCF (Chandler, 2003; Hartshorn & Evans, 2015), yet earlier work (Robb et al., 1986) found no significant effects. This variability underscores the intricate interaction between feedback type, learner engagement, and linguistic development. Importantly, ecological validity must be considered to ensure that findings from controlled studies are applicable to authentic learning environments, ultimately shaping the real-world effectiveness of both AI- and teacher-mediated feedback.

Simultaneously, the rapid advancement of AI-driven chatbots has opened new avenues for exploring their role in delivering WCF in L2 writing instruction (Barrot, 2023; Pecorari, 2023; Praphan & Praphan, 2023; Shaofeng, 2025; Warschauer et al.,

2023). Equipped with human-like conversational abilities, AI chatbots offer unprecedented efficiency and accessibility, potentially overcoming some limitations of traditional AWE systems, which have been criticized for formulaic feedback focused mainly on surface errors (Chen & Cheng, 2008; Li et al., 2015). Emerging studies indicate that AI-generated feedback can match or even surpass human instructors in certain feedback domains (Bouziane & Bouziane, 2024; Escalante et al., 2023; Steiss et al., 2024).

The literature on AI in L2 writing predominantly employs heuristic approaches to assess the affordances of AI chatbots in providing WCF (see [Supplementary Appendix B](#)). Key themes have emerged: the unique features of AI-generated feedback—such as its volume and quality—and its capacity to facilitate meaningful learner development. Many investigations use comparative designs to evaluate AI feedback relative to human teacher feedback.

Regarding the distinctive characteristics of AI-generated WCF, recent research highlights several advantages. AI chatbots tend to provide more extensive feedback than human teachers (Guo & Wang, 2024; Wang et al., 2024) and deliver feedback more evenly across content, language, and organizational aspects (Guo & Wang, 2024). They also exhibit higher readability levels (Dai et al., 2023; cf. Guo & Wang, 2024). For instance, Wang et al. (2024) found ChatGPT's accuracy in identifying argumentative essay components reached 91.8%, closely mirroring human evaluations, and in some cases, such as argument claims, achieved complete agreement (100%) with human instructors. However, Steiss et al. (2024) observed that ChatGPT outperformed humans only when provided with explicit rubric criteria, while human instructors offered superior feedback quality in other categories.

From a pedagogical standpoint, the instant availability of comprehensive, accurate, and accessible AI-generated feedback can considerably reduce instructors' workload. Nevertheless, AI feedback sometimes lacks focus, tends to be verbose or redundant (Guo & Wang, 2024; Lin & Crosthwaite, 2024), and may be perceived as less engaging by learners, potentially affecting motivation and the writer's sense of ownership (Escalante et al., 2023; Huang & Deng, 2025; Praphan & Praphan, 2023). Such drawbacks underscore the importance of integrating AI feedback thoughtfully within instruction.

Further studies have explored how human teachers and AI chatbots differ in feedback priorities and strategies. Steiss et al. (2024) report that human instructors excel at providing clear improvement directives, prioritizing critical writing features, and adopting supportive tones. Guo and Wang (2024) found that while human teachers focus more on content- and language-related issues, AI feedback is more balanced across content, language, and organization. Moreover, teachers employ a mix of directive and informative feedback for content issues and combine informative and query-based feedback for language, whereas ChatGPT favors directive and summary feedback for organization and mostly directive feedback for language problems. Lin and Crosthwaite (2024) highlight that teachers use both direct and indirect WCF frequently, but ChatGPT rarely provides indirect feedback, instead relying on metalinguistic explanations and reformulations. Notably, when ChatGPT does use indirect feedback, it focuses on local errors, contrasting with teachers who apply indirect feedback primarily to global writing issues.

Given that WCF priorities and delivery methods significantly influence L2 writing development, ongoing research is critical to understand their distinct effects. While AI-generated feedback offers scalability and immediacy, its integration alongside human pedagogical expertise is essential to maximize benefits for learners' writing performance.

From AI feedback to teacher WCF: comparing pedagogical effectiveness

While currently available reports have clarified the capabilities and limitations of AI chatbots in comparison to human writing instructors, a key question that warrants further investigation is the effectiveness of these tools in enhancing L2 learners' writing outcomes. After all, fostering meaningful progress in learner writing is a primary goal of WCF and a crucial component of any writing intervention. Given this, the second theme emerging in the literature focuses on how AI-generated WCF affects L2 writing outcomes. Related to this, Escalante et al. (2023) investigated whether the application of AI-generated feedback led to enhanced writing performance among English as a New Language (ENL) students compared to feedback from a human tutor. Their findings indicated no significant advantage of AI-generated feedback over human tutor feedback in promoting linguistic development. In contrast, Wiboolyasarin et al. (2024) investigated the impact of a collaborative writing intervention supported by AI feedback on learners' Thai writing proficiency. Their study found statistically significant improvements in writing quality for the experimental group, with a large effect size, while the control group showed no significant pre- and post-test differences. These contrasting results highlight the complexity of L2 writing outcomes associated with AI-generated WCF, suggesting that while some contexts may benefit from AI feedback, its effectiveness can vary depending on instructional approaches and learner demographics.

While the emerging literature, as discussed earlier, provides valuable insights into the distinctive features of AI-generated WCF and its influence on writing development, it is essential to connect these new insights with the well-established body of research on teacher-generated WCF. This line of inquiry is critical for several reasons. First, as AI-powered feedback is a recent innovation, evaluating its capabilities in comparison to human-provided WCF can help determine its appropriate role in current and future L2 writing classrooms. For decades, teachers have been the primary source of WCF, and any move to outsource part of this process to AI tools must be supported by evidence of their pedagogical effectiveness.

Moreover, practical constraints—such as large class sizes, limited instructional time, and resource shortages—often force teachers to compromise the quantity and quality of feedback they can offer. Teachers, despite their pedagogical expertise, may struggle to provide individualized, timely feedback to all students. AI-powered tools, in contrast, offer the potential to address some of these limitations by delivering feedback that is scalable, timely, and potentially personalized. However, realizing these benefits depends on a clear understanding of how AI-generated WCF compares to traditional teacher feedback. The following section reports key themes in traditional WCF research, offering a basis for contextualizing the pedagogical implications of AI in L2 writing instruction.

The present study

While prior research has highlighted the potential benefits of AI-generated feedback, empirical evidence directly comparing its effectiveness to teacher-provided feedback remains limited. Furthermore, existing studies (Escalante et al., 2023; Wiboolyasarin et al., 2024) often lack specificity in how effectiveness is measured, relying on broad and loosely defined categories, such as writing quality, linguistic progress, content, and organization. This methodological ambiguity raises concerns about the reliability and validity of their findings.

At the same time, although AI shows considerable promise, scholars caution against positioning it as a replacement for human instructors (Cardona et al., 2023). Instead, AI is best viewed as a complement to established pedagogical approaches, with teachers continuing to play a central role in instruction. Current recommendations for automated writing evaluation systems emphasize their optimal use in conjunction with human feedback (Link et al., 2022)—a principle that extends to AI-mediated WCF. Identifying where AI tools are most and least effective can help educators integrate them more thoughtfully, maximizing benefits while mitigating potential drawbacks. Against this backdrop, the present study addresses a timely and important niche by empirically examining the comparative effectiveness of AI-mediated and teacher-mediated WCF in L2 narrative writing. Effectiveness is operationalized through measurable gains in accuracy, as well as lexical and syntactic complexity (see Section Data analysis). The study addressed three research questions (RQs):

- RQ1: How do AI-mediated and teacher-mediated WCF affect accuracy in L2 narrative writing?
- RQ2: How do AI-mediated and teacher-mediated WCF affect lexical and syntactic complexity in L2 narrative writing?
- RQ3: How do AI-mediated and teacher-mediated WCF affect quality scores in L2 narrative writing?

Methods

Participants

This study included a total of 169 first-year ESL undergraduate participants (73 men and 96 women) from 12 intact classes at a major university in the US. Of these 12 classes, five classes with 80 students received teacher-mediated feedback, while seven classes with 86 students received AI-mediated feedback. Three students—one from the teacher-mediated group and two from the AI-mediated group—were excluded from the analysis due to missing revision submissions, resulting in a final sample of 166 participants. In terms of demographics, participants ranged in age from 18 to 20 years ($M=18.5$ years, $SD=3.2$). On average, participants had studied English for 4.3 years ($SD=2.4$) and had completed ~ 3.2 writing courses ($SD=1.1$) during their education. The sample was internationally diverse, with 29.6% from China ($n=50$), 24.9% from South Korea ($n=42$), 14.8% from Brazil ($n=25$), 11.4% from India ($n=19$), 10.2% from Vietnam ($n=17$), and 7.7% from France ($n=13$). All participants were homogeneous regarding English writing proficiency, as evidenced by their

International English Language Testing System (IELTS) Task 2 writing section, which ranged from 5.0 to 6.0—equivalent to a B2 level on the CEFR scale.

Writing task

For this study, participants completed a written literacy narrative in response to the prompt: *Reflect on your experience as a writer and describe the events and individuals that have influenced your attitudes and beliefs about reading and writing.* The genre was chosen because it is a foundational genre in first-year ESL writing curricula, helping students develop coherence, organization, and sentence structure (Abdi Tabari, 2023). By fostering personal expression, literacy narratives also promote engagement and motivation, key factors in language learning. Additionally, narrative writing provides an invaluable opportunity to analyze both writing development and the effectiveness of WCF, as the inclusion of vivid, personal details generates rich data for assessing L2 writing performance.

Intervention

The intervention was implemented over two consecutive weeks and followed a structured three-stage writing cycle designed to integrate task completion, feedback, and revision within the regular class schedule. In Week 5, students received verbal instructions and completed the original writing task. In Week 6, a one-time orientation session was conducted to prepare students for the responsible use of AI-generated feedback, followed by the delivery of individualized feedback and an immediate revision session. Table 1 summarizes the timing, activities, and student expectations for each stage of the intervention.

This study examined the comparative effects of teacher- and AI-mediated WCF on three key dimensions of L2 narrative writing: accuracy, linguistic complexity, and overall quality. Participants were drawn from intact class sections and assigned to one of two treatment groups based on their course enrollment. Group 1 (teacher-mediated WCF) included five classes ($n=80$), where students received individualized feedback from their instructor. Group 2 (AI-mediated WCF) included seven classes ($n=86$), where feedback was generated by a generative AI tool (ChatGPT-4). Three students—one from the teacher-mediated group and two from the AI-mediated group—did not submit revised drafts and were excluded from the analysis, yielding a final sample of 166 participants.

All courses were taught by three experienced instructors who followed a shared syllabus, used identical instructional materials, and adhered to the university's Core Writing program curriculum. Each class met once weekly for two and a half hours.

In the teacher-mediated WCF group, instructors provided feedback using a standardized rubric developed by the Core Writing planning team. While instructors were not limited to fixed error categories, the rubric ensured consistent attention to grammar, vocabulary, organization, and content. Feedback was delivered electronically *via* the university's course management system (Canvas) as typed comments within the document margins. Feedback emphasized clarity, accuracy, and cohesion. Sample teacher-mediated feedback is provided in [Supplementary Appendix C](#).

Table 1. Overview of the three-stage intervention procedure, including timing, activities, and student expectations.

Stage	Timing	Activities and expectations
Stage 1: Writing task	Week 5	At the beginning of class, students received 15 min of verbal instructions on the writing task and its requirements. Students then completed an in-class narrative writing task responding to a guided prompt under timed conditions, producing a complete draft demonstrating their narrative skills.
Stage 2: Orientation session	Week 6 (before feedback session)	Students attended a single 30-min in-class orientation focused on prompt literacy and responsible use of AI-generated feedback. The session introduced how large language models work, emphasized critical evaluation of AI feedback, and guided students to use AI feedback reflectively and ethically.
Stage 3: Feedback and revision	Week 6 (same class session as orientation)	Students received individualized written corrective feedback—either from their instructor (teacher-mediated group) or the AI tool (AI-mediated group). They revised their drafts during the same session and submitted the final versions online.

In the AI-mediated WCF group, feedback was generated by ChatGPT-4. Each student's full draft was submitted to the AI using a standardized prompt developed by the research team to elicit criterion-based feedback aligned with the Core Writing rubric. The AI feedback addressed five core evaluation criteria: focus on the prompt, use of narrative elements beyond plot, organization based on the five stages of plot, word choice including sensory details and figurative language, and writing conventions, such as grammar and mechanics. Sample AI feedback is included in [Supplementary Appendix C](#). A zero-shot prompting approach was used, providing structured instructions without specific examples to ensure rubric alignment while maintaining generalizability.

To ensure ethical and responsible AI use, the 30-min orientation session before the feedback/revision phase was held once in Week 6 and attended by all students in the AI-mediated group. This session highlighted critical evaluation strategies to prevent overreliance on AI and encouraged reflective engagement with the feedback. Instructors monitored revision processes and provided support as needed to maintain academic integrity.

Students were not required to generate their own AI prompts; instead, a standardized input format was applied to all drafts to maintain consistency. The feedback process was designed to reflect authentic pedagogical and technological practices by not restricting feedback focus to specific linguistic features. The one-week interval between the initial writing and feedback/revision session allowed sufficient time for feedback preparation and student reflection while fitting into the regular course schedule. Due to classroom constraints, no no-feedback control group was included. However, this design allowed a controlled comparison of teacher- and AI-mediated WCF effects on L2 narrative writing development.

AI tool and prompt design

To implement AI-mediated WCF, this study utilized ChatGPT-4 (developed by OpenAI), accessed by the instructor group *via* the official platform at the time of data collection. ChatGPT-4 was selected due to its advanced linguistic capabilities

and widespread accessibility in educational contexts. Prompting for the AI group was carefully designed to align with current best practices in prompt engineering (Giray, 2023) and principles of zero-shot reasoning (Kojima et al., 2022), which allow large language models (LLMs) to generate context-sensitive responses without task-specific fine-tuning or in-context examples.

A zero-shot prompting approach was adopted, in which the model received structured instructions without being shown example inputs or outputs. This design choice was guided by emerging evidence that, with detailed instructions, LLMs, such as ChatGPT-4 can reason and generate accurate feedback in unfamiliar tasks (Brown et al., 2020; Kojima et al., 2022). Each student's unedited narrative draft was input into the model along with a standardized feedback prompt created by the research team. The AI prompt instructed the model to evaluate the narrative writing based on the five key rubric categories used in this study:

1. Focus on Prompt, assessing how clearly and fully the essay addressed and developed the assigned topic;
2. Narrative Elements (other than plot), evaluating the effective use of techniques, such as character, setting, dialogue, and conflict;
3. Five Stages of Plot (Organization), assessing the presence and coherence of exposition, rising action, climax, falling action, and resolution;
4. Word Choice, focusing on the use of vivid sensory details, figurative language, and varied sentence structures to establish voice and imagery; and
5. Writing Conventions, including grammar, mechanics, and usage.

All AI-generated feedback was reviewed by the first author before distribution to ensure alignment with instructional goals and consistency in rubric interpretation. This process allowed for the delivery of individualized, rubric-based, and criterion-referenced feedback while minimizing researcher involvement in content generation. The full version of the standardized AI prompt is provided in [Supplementary Appendix D](#) to promote transparency and replicability.

Data analysis

In the present study, the quantitative patterns of accuracy, complexity, and quality measures were inferred using linear mixed-effects modelling (LMM; fixed effect: group and version, random effect: participant). The metrics used were those commonly used in the field of L2 writing research: (1) for accuracy, we used the metric of ratio of error-free clauses (EFC/C); (2) for lexical complexity, we used measure of textual lexical diversity (MTLD, cf. McCarthy & Jarvis, 2010), log-transformed average word frequency (LogWF), lexical density or the ratio of content words (Density), and mean lexical familiarity for measuring psycholinguistic property; (3) for syntactic complexity, we used T-unit based measures from the L2SCA analyzer framework (Lu, 2017) including mean length of T-unit (MLT) for measuring overall complexity, dependent clauses per T-unit (DC/T) for measuring subordination, coordinated phrases per T-unit (CP/T) for measuring coordination, and complex nominals per T-unit (CN/T) for measuring the complexity of core argument

structures (e.g. subjects and objects). Finally, for textual quality, we used rating scores based on prompt focus, narrative elements, staging, word choices, and overall quality, see above. The correlation of the metrics in each domain is summarised in [Supplementary Appendix E](#).

All essays were analyzed using both automated and manual procedures. Linguistic complexity was measured with automated tools, such as TAALES and L2SCA, which have demonstrated high reliability (see Lu, 2017); thus, inter-rater reliability was not assessed for these metrics. In contrast, accuracy (EFC/C) and L2 writing quality were manually coded by the first author. To ensure reliability, 50% of the essays were independently re-coded by a trained colleague with expertise in L2 writing assessment. Both raters had prior experience with analytic rubrics and aligned their interpretations through preliminary discussions. Inter-rater reliability, assessed *via* Pearson's *r*, showed strong agreement: EFC/C (.92), Focus on Prompt (.88), Narrative Elements (.94), Organization (.90), Word Choice (.86), and Writing Conventions (.85), supporting the consistency of the scoring.

Since our study has a clear 2-by-2 design (AI *vs.* Teacher-mediated feedback, first *vs.* revised draft), we used *F*-tests to determine the effect of group or feedback mediator, draft version, and any interaction effects. Since the participants in the two groups had similar English proficiency levels, we did not expect substantial differences based on group alone. Regarding the draft version, we expected significant effects; otherwise, the feedback did not translate into textual improvement after revision. Lastly, regarding the interaction between group and version, any significant effect would indicate a mediator-specific difference after the revision, thus eliciting any AI-advantage or Teacher (Human) advantage. We use marginal *R*² as a measure of goodness-of-fit for each LMM, and the *R*² value is analogous to eta-squared in traditional ANOVA models, indicating the proportion of variance explained by the whole model. According to the field-specific yardsticks for interpreting effect sizes (Plonsky & Oswald, 2014), small, medium, and large effects correspond to Cohen's *d* values at 0.4, 0.7, and 1.0. Based on the mathematical relation between *d* and *R*² (i.e. $R^2 = d^2 / (d^2 + 4)$), we used benchmarks at 0.038, 0.109, and 0.200 to determine a small, medium, and large effect.

Results

L2 writing accuracy

First, we examined the accuracy of L2 writing measured by *the ratio of error-free clauses to all clauses* (EFCs/C) before and after revision based on WCF in both groups. As [Figure 1](#) shows, the two groups had similar accuracy levels in the first draft, and both groups showed a clear improvement in their revised essays. When checked by *F*-tests based on linear mixed-effects modeling (LMM, see [Table 2](#)), there was no significant effect of Group. This was expected mostly due to the groups' comparable proficiency levels. A significant effect for Version ($p < .001$) and a significant interaction between Group and Version ($p < .001$) were observed, indicating a Group-specific improvement pattern. In this case, participants receiving teacher-mediated WCF showed a stronger trend in accuracy improvement than those receiving AI-mediated

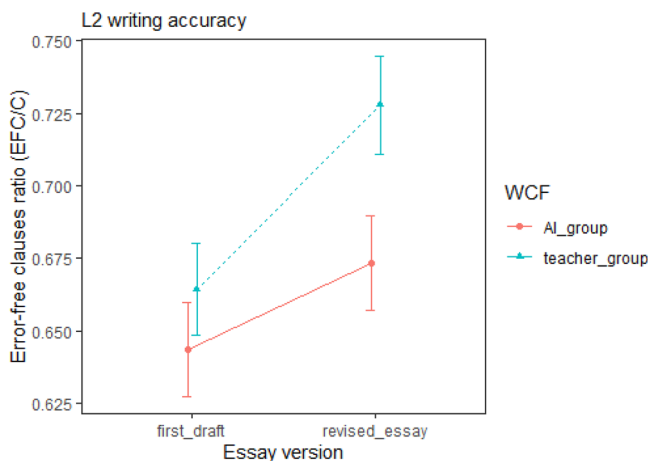


Figure 1. L2 writing accuracy before and after receiving WCF in two groups. Error bars indicate standard errors.

WCF. To summarise, both teacher and AI groups showed significant improvements in accuracy, but teacher-mediated WCF was more advantageous than AI-mediated WCF for accuracy improvement (although the effect sizes were mostly small).

Lexical complexity metrics

Next, we gauged four lexical complexity metrics: *Measure of textual lexical diversity (MTLD)*, *lexical sophistication (LogWF)*, *lexical density (content word ratio)*, and *lexical familiarity (MRC familiarity)*. The descriptive results are presented in Figure 2, and the inferential statistics are shown in Table 2. As a general trend, the Group effect was not significant for all lexical metrics ($p's > .05$), suggesting that the two groups again had similar baseline performance levels; at the same time, the Version effect was significant for all four metrics ($p's < .001$), suggesting that all participants in both groups showed a significant change in the lexical metrics, including increased MTLD and lexical diversity and decreased LogWF and MRC familiarity. All these changes suggest improved lexical complexity (Note that lower word frequency and familiarity imply more advanced lexical usage). Significant interactions were detected in MTLD and LogWF ($p's < .001$). In each case, the effect suggested an advantage of teacher-mediated WCF over the AI group, as the lines were steeper in the former. It should be noted that the effect sizes for the three measures of lexical complexity were small, except MTLD, which showed a medium effect size (see Table 2). To summarise, L2 writers in both groups showed increased lexical complexity after revision based on WCF, but teacher feedback has a further advantage in lexical diversity and sophistication over AI feedback.

Syntactic complexity metrics

We also measured syntactic complexity using *mean length of T-units (MLT)*, *dependent clauses per T-unit (DC/T)*, *coordinate phrases per T-unit (CP/T)*, and *complex*

Table 2. Inferential statistics for accuracy, complexity, and quality measures in two groups (teacher group vs. AI group) and two essay versions (first drafts vs. revised essays).

Measure	Group		Version		Group × Version		R_m^2
	F	p	F	p	F	p	
1. Accuracy							
EFC/C	2.71	.102	490.8	<.001***	64.9	<.001*** ^T	0.042 ^S
2. Lexical							
MTLD	1.1	.303	90.0	<.001***	17.8	<.001*** ^T	0.131 ^M
LogWF	0.3	.596	80.6	<.001***	11.5	<.001*** ^T	0.075 ^S
Density	0.3	.575	30.4	<.001***	3.4	.069	0.026 ^S
Familiarity	0.3	.603	45.4	<.001***	3.5	.062	0.043 ^S
3. Syntactic							
MLT	0.3	.582	54.2	<.001***	7.2	.008** ^{AI}	0.050 ^S
DC/T	0.04	.848	39.1	<.001***	5.4	.201	0.047 ^S
CP/T	0.7	.405	16.8	<.001***	3.7	.056	0.015 ^S
CN/T	0.5	.488	34.9	<.001***	10.4	.002** ^{AI}	0.027 ^S
4. Quality							
Prompt	3.6	.058	50.6	<.001***	0.1	.714	0.110 ^M
Narrative	0.5	.490	81.9	<.001***	2.6	.111	0.158 ^M
Stages	0.03	.871	57.9	<.001***	0.5	.467	0.106 ^M
Word choice	3.1	.078	46.8	<.001***	8.6	.004** ^T	0.143 ^M
Conventions	47.9	<.001***	113.7	<.001***	2.6	.107	0.292 ^L
Overall	4.1	.045*	223.5	<.001***	0.5	.501	0.336 ^L

Note. * $p < .05$, ** $p < .01$, *** $p < .001$. Small, medium, and large effects are determined at 0.038, 0.109, and 0.200 benchmarks.

^TAdvantage of the teacher group.

^{AI}Advantage of the AI group.

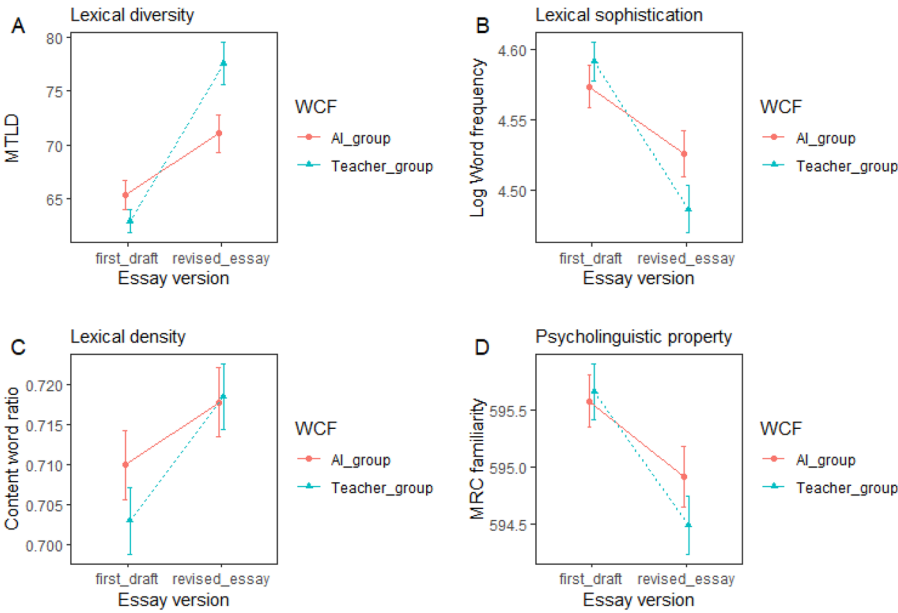


Figure 2. Lexical complexity metrics of L2 writing before and after written corrective feedback in two groups. Error bars represent standard errors.

noun phrases per T-unit (CN/T). As with lexical metrics, we presented the mean values of syntactic metrics in line graphs in Figure 3, and the inferential statistics are available in Table 2 above. First, the Group effect was again not significant for

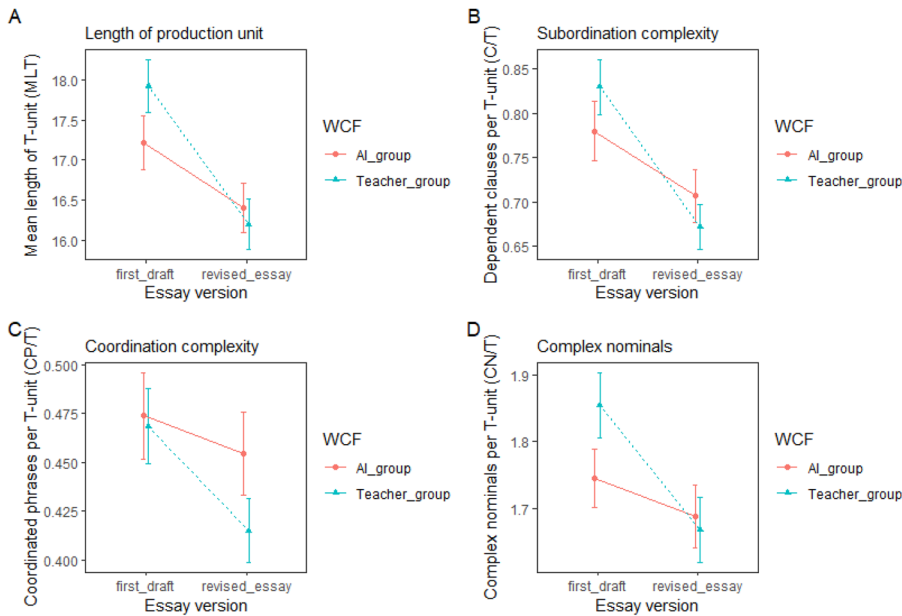


Figure 3. Syntactic complexity metrics of L2 writing before and after written corrective feedback in two groups. Error bars represent standard errors.

all metrics, as expected. However, this time we observed that all four syntactic metrics showed a drop in the revised narrative essays (p 's < .001), suggesting that the syntactic complexity level *decreased* after the revision, regardless of the mediator of WCF: L2 writers tended to produce shorter T-units (MLT), use less subordination (DC/T), coordination (CP/T), and complex nominals (CN/T). This pattern contrasts sharply with the lexical complexity metrics, where a consistent improvement is observed. In other words, our data suggested a *trade-off* between lexical and syntactic complexity as an effect of WCF, and the pattern was largely consistent between the teacher group and the AI group. Additionally, significant interaction effects were detected for MLT and CN/T (p = .008 and .002, respectively). In this case, we found an AI advantage since the decrease was more evident in L2 writers who received teacher-mediated WCF, as indicated by steeper lines in the graph. The observed statistical significance for all of the measures was, however, small (see Table 2).

L2 writing quality scores

Regarding the holistic ratings of the narrative writing performance, we also included analyses on five quality dimensions (*prompt focus, narrative components, stage management, word choices, and writing conventions*) as well as the averaged, overall quality measure (see Figure 4). Again, the inferential statistics are available in Table 2. Here, the Group effect was not significant in most of the metrics as expected, except for conventions and the overall quality score (p < .001 and p = .045, respectively), representing uncontrollable data variance from a classroom-based design. Nevertheless, the effect of Version was significant in all quality dimensions (p 's < .001),

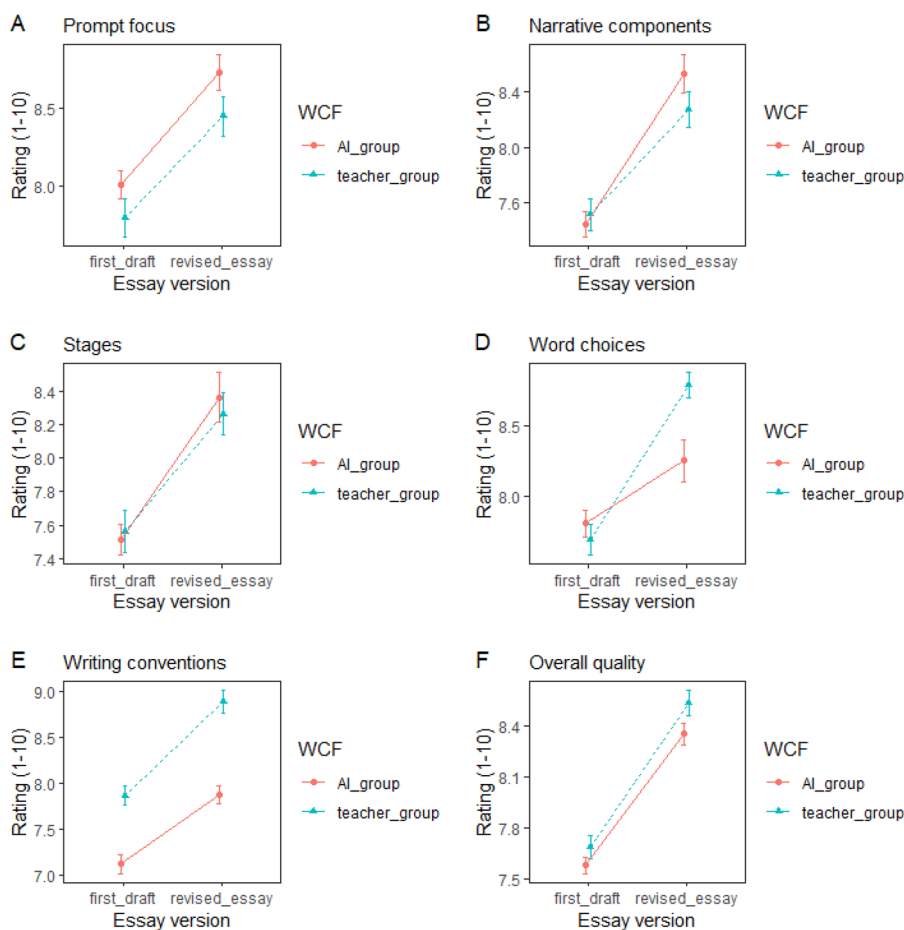


Figure 4. Quality scores of L2 narrative writing before and after written corrective feedback in two groups. Error bars represent standard errors.

suggesting that the revised essays had better narrative writing quality scores. Significant interaction was only detected in word choices ($p = .004$), and in this case, there was a steeper increase in the teacher group over the AI group, suggesting an advantage of teacher-mediated WCF. The interaction effects in other dimensions were not significant, suggesting that the improvement was comparable between teacher- and AI-mediated WCF for prompt focus, narrative components, stage management, writing conventions, and overall quality. Effect sizes for prompt focus, narrative components, stage management, and word choices were medium, whereas writing conventions and overall quality showed large effect sizes (see Table 2).

To summarise, an advantage for teacher-mediated WCF was supported by accuracy (EFC/C), lexical diversity (MTLD), lexical sophistication (LogWF), and word choices rating; an advantage of automatized, AI-mediated WCF was supported by two syntactic metrics of production unit length (MLT) and complex nominal usage (CN/T) (less decreases, not more improvements). No clear advantage was detected in lexical density, familiarity, coordination, subordination, or most narrative writing quality

ratings. Given the hypothesized trade-off between lexical and syntactic complexity measures, the current study suggests that WCF has an effect of shifting the L2 writer's attention to use more advanced lexical resources at the cost of reducing the syntactic complexity. This trade-off seems to be more pronounced with traditional teacher-mediated WCF over automated AI-mediated WCF, although both groups showed consistent patterns overall.

Discussion

This study investigated the effects of AI-mediated and teacher-mediated WCF on L2 narrative writing regarding accuracy, lexical and syntactic complexity, and holistic quality. The findings offer differentiated insights for each research question, each of which merits in-depth consideration.

RQ1: How do AI-mediated and teacher-mediated WCF affect accuracy in L2 narrative writing?

The findings demonstrate that both AI- and teacher-mediated WCF significantly enhanced learners' grammatical accuracy in L2 narrative writing. However, teacher-mediated feedback yielded a more substantial improvement, highlighting the enduring pedagogical value of human feedback in writing instruction. This outcome corroborates earlier studies (e.g. Steiss et al., 2024), which have consistently shown higher accuracy scores when students receive feedback from teachers rather than from AI systems.

Although ChatGPT-generated WCF led to statistically significant accuracy gains, its relatively smaller effect size points to a fundamental pedagogical distinction. Feedback effectiveness is not solely determined by the ability to detect and correct grammatical errors but by how feedback shapes learners' cognitive engagement, noticing, and internalization. Teacher-mediated WCF typically includes metalinguistic explanations, clarification requests, and prompts tailored to learners' individual needs (Bitchener & Knoch, 2010). These features promote deeper syntactic restructuring and facilitate long-term improvements in linguistic competence. In contrast, AI-generated feedback tends to be more directive and generalized, often offering surface-level suggestions, such as word replacements, deletions, or insertions (Guo & Wang, 2024; Lin & Crosthwaite, 2024). While such feedback is often accurate and immediate, it may lack the instructional specificity and developmental sensitivity needed to trigger deeper learning.

Another key strength of teacher-mediated feedback lies in its adaptive and interactive nature. Teachers can interpret learners' error patterns, respond flexibly, and scaffold feedback within the learner's Zone of Proximal Development (ZPD), fostering not just correction but development (Thorne & Lantolf, 2007). These dialogic features are central to the effectiveness of feedback as a learning tool, particularly in contexts where language development is shaped by the interaction and negotiation of meaning. Recent studies (e.g. Escalante et al., 2023; Peng et al., 2023) suggest that learners associate teacher feedback with greater cognitive involvement, reporting that it prompts reflection, deeper processing, and sustained engagement with form.

Metacognitive engagement is also more likely to be elicited by teacher feedback. Guo and Wang (2024), for instance, found that teacher comments were significantly more informative and inquiry-driven than those generated by ChatGPT. Rather than simply prescribing corrections, teachers often prompt learners to diagnose the nature of their errors, reflect on possible alternatives, and revise with intention. This level of engagement can lead to greater internalization of grammatical rules and more meaningful revisions. In contrast, AI-generated feedback, while efficient, may promote more superficial error correction and limit opportunities for learner-driven problem-solving (Fokides & Peristeraki, 2025).

In sum, while AI-mediated WCF is a promising tool—particularly valued for its scalability, speed, and linguistic precision—it currently lacks the pedagogical intentionality and contextual responsiveness that characterize expert teacher feedback. The greater gains observed in the teacher-mediated condition in this study underscore the view that effective written corrective feedback is not merely a matter of error detection but a pedagogical act grounded in interaction, scaffolding, and learner development. These findings support the growing consensus (e.g. Lundberg, 2024; Nassaji, 2020) that AI can complement but not replace the complex instructional functions of human feedback in L2 writing instruction.

RQ2: How do AI-mediated and teacher-mediated WCF affect lexical and syntactic complexity in L2 narrative writing?

The study found significant gains in lexical sophistication for both groups, particularly in measures, such as MTLT and the use of lower-frequency vocabulary (LogWF). The teacher-mediated group demonstrated larger gains in these dimensions, suggesting that teacher feedback more effectively promoted lexical growth. Teachers often use contextualized lexical suggestions and encourage learners to paraphrase or use synonyms, which not only help expand their vocabulary but also foster deeper cognitive processing (Abdi Tabari et al., 2023). As Storch and Wigglesworth (2010) found, feedback that encourages learners to think critically about their word choices promotes deeper learning and retention. By providing WCF on lexical choices, teachers help learners refine their vocabulary usage in ways that transfer across L2 writing tasks, leading to greater lexical diversity and sophistication.

However, the results for syntactic complexity showed a general decline across all measures (MLT, DC/T, CP/T, CN/T), indicating that learners may have simplified their syntax during revision. This trend aligns with Skehan's (2014) Limited Attentional Capacity Model, which suggests that learners prioritize accuracy or lexis when cognitive resources are stretched, often at the expense of syntactic complexity. This trade-off between accuracy and complexity is well-documented in L2 writing research, which posits that learners tend to focus on one aspect of language (e.g. grammar) at the expense of others (e.g. sentence complexity) when under cognitive load (Silva et al., 2024). One possible explanation for this pattern is that lexical revisions are generally more manageable than syntactic ones. Lexical changes often involve substituting individual words or short phrases, whereas syntactic revisions typically require broader structural adjustments and greater control over sentence-level

organization (Maamuujav et al., 2021). As such, the observed increase in lexical sophistication in both groups' revisions may reflect a strategic preference for revisions that are cognitively less demanding.

Interestingly, the AI-mediated group exhibited smaller declines in MLT and CN/T, suggesting that the cognitive load imposed by AI feedback was potentially lower. AI feedback, with its rapid and visually salient corrections, may have allowed learners to devote more of their cognitive resources to maintaining syntactic complexity. In contrast, teacher feedback might have required learners to process more detailed explanations, resulting in more cognitive strain and a consequent reduction in syntactic complexity. This finding invites a reconsideration of the role AI can play in L2 writing: despite its lack of pedagogical nuance, its automated and consistent nature may support sustained syntactic output under time pressure or cognitive load. AI feedback might also be perceived as less intrusive, enabling learners to focus more on sentence-level fluency and structural complexity without feeling overwhelmed (Plate et al., 2024). This suggests that AI feedback can complement teacher feedback, particularly in contexts that require rapid production or multiple drafts, as AI tools can help learners maintain syntactic complexity while focusing on form.

RQ3: How do AI-mediated and teacher-mediated WCF affect quality scores in L2 narrative writing?

Holistic quality ratings improved significantly for both groups, indicating the positive effects of WCF on overall L2 writing quality. Teacher-mediated feedback led to greater improvements in the lexical choice dimension of quality, further supporting the notion that teachers are more effective at guiding learners toward contextually appropriate and precise vocabulary. This aligns with Ferris's (2006) assertion that teachers are uniquely equipped to support lexical development through selective, meaningful interventions that promote the use of accurate and contextually appropriate vocabulary.

The general quality gains across both groups may be attributed to the fact that both types of WCF triggered revisions at multiple discourse levels. Learners in both groups were likely exposed to feedback addressing issues, such as organization, cohesion, and idea development, which likely stimulated macro-level revisions. This corroborates earlier research suggesting that WCF, regardless of source, can facilitate global improvements when learners are encouraged to revisit content and structure (Leow et al., 2022). While the provision of comprehensive feedback by teachers and its effectiveness is well documented (Griffiths et al., 2016; Parr & Timperley, 2010), emerging research indicates that AI chatbots also possess the capability to deliver such feedback (Wang et al., 2024). The holistic nature of both teacher and AI feedback might have encouraged learners to improve their writing not just at the level of syntax and vocabulary but also in terms of overall coherence and organization. This interpretation aligns with emerging scholarship suggesting that AI-generated WCF can be as effective as teacher feedback in meaningfully enhancing students' writing performance. For instance, Wiboolyasarin et al. (2024) documented

significant gains in learners' writing quality, particularly in areas, such as structure, word count, and language use, following the integration of ChatGPT into the feedback process. This is a notable development, especially considering earlier critiques of AWE systems, which often emphasized their limited capacity to move beyond surface-level corrections (Chen & Cheng, 2008; Li et al., 2015; Zhang, 2020).

Interestingly, the absence of significant interaction effects in dimensions like organization and coherence suggests that both types of feedback contributed equally to structural improvements. It is plausible that both AI and teacher comments provided salient cues for discourse-level changes, whether through direct suggestions (in the teacher group) or error pattern highlighting (in the AI group). Thus, AI feedback might indirectly support higher-order writing development, even without explicit instruction. This highlights the utility of AI feedback as a tool for reinforcing larger structural issues, even when it lacks the direct pedagogical expertise of a human instructor.

Conclusion

WCF is a cornerstone of effective L2 writing instruction. Yet, providing high-quality feedback is challenging, particularly in classrooms with large teacher-student ratios, which often force instructors to compromise on feedback focus and strategy. The advent of generative AI offers promising support in mitigating some of these challenges. Large language models (LLMs) like ChatGPT, trained on expansive and diverse textual corpora, can generate feedback that goes beyond surface-level issues, as demonstrated by our findings. This is a major development considering automated writing evaluation systems' challenges with providing comprehensive, discourse-level feedback (Chen & Cheng, 2008; Li et al., 2015). Moreover, the immediacy of AI feedback can facilitate iterative drafting and sustain learner engagement, especially in contexts where the quantity and quality of teacher feedback are constrained by time or class size. This scalability makes AI feedback a valuable supplement in resource-limited educational settings.

Moreover, unlike human instructors, who may vary in expertise across vocabulary knowledge, grammar, discourse organization, or genre conventions, AI systems can draw on a broad base of knowledge to offer feedback across multiple dimensions. While AI-generated feedback has its limitations, such as the potential for inaccuracies or redundancies, its breadth ensures learners receive at least preliminary guidance on areas that might otherwise be overlooked by human teachers. That said, using AI as a feedback tool on student essays should complement, not replace, a teacher's pedagogical practices—much like dietary supplements are designed to enhance a balanced diet, not substitute real food. Just as relying solely on supplements would neglect the richness and complexity of whole meals, depending solely on AI for feedback risks overlooking the nuanced, human-centered support that students need to grow as writers.

Yet, the impact of AI-generated feedback, including WCF, should be interpreted with caution. Comparing AI feedback to teacher feedback is not a direct and fair comparison between equivalent sources. Instead, it contrasts an idealized, fatigue-free system with instructors operating under real-world constraints—limited time, large

classes, cognitive fatigue, and competing responsibilities, to name a few. AI can generate unlimited, immediate, and standardized feedback without being affected by such challenges. This fundamental difference raises important questions about the practical use of AI-mediated WCF in classroom settings. While AI feedback may rival or even surpass teacher feedback under certain controlled conditions, it cannot replicate the pedagogical nuance, individualized soft-tuning, or relational elements of teacher-student interaction. Thus, AI-mediated WCF should be seen as a complementary tool, particularly given its susceptibility to generating inaccurate feedback. This concern aligns with Steiss et al. (2024) call for a *utility-alongside-fallibility* perspective, emphasizing the need for teachers to critically evaluate and supplement AI-generated feedback to ensure pedagogical soundness.

Given these considerations, it is essential to explore how the complementary strengths of both AI and teacher-mediated feedback can be harnessed in pedagogically sound ways. This study offers several practical implications for language educators and curriculum designers seeking to optimize feedback practices in L2 writing instruction. First, teacher-mediated WCF remains critical for promoting grammatical accuracy and lexical sophistication. Its nuanced, individualized, and context-sensitive nature supports deeper language processing and sustained linguistic development. Instructors should therefore remain actively involved in providing feedback, especially during later stages of writing when targeted revisions are most impactful. Second, AI-mediated WCF offers value through its immediacy, consistency, and efficiency. It supports learners in sustaining syntactic complexity and addressing lower-order concerns during early drafting. Tools like Grammarly or Write & Improve can be integrated as supplemental feedback mechanisms, particularly in large classes where individualized feedback is limited. Third, the study supports a hybrid feedback model in which AI and teacher feedback serve complementary roles. Teachers can sequence feedback strategically—using AI to address surface-level and fluency issues in initial drafts, followed by teacher feedback to tackle higher-order concerns, such as lexical precision, cohesion, and content development. This staged approach fosters iterative refinement and supports learners through multiple levels of textual improvement. Finally, instructors should offer metacognitive training to help learners interpret and use both types of feedback effectively. Students need guidance in critically engaging with AI suggestions and in prioritizing teacher feedback to support meaningful revision rather than superficial edits. This pedagogical support can help maximize the benefits of each feedback modality.

While this study contributes to the growing literature on WCF in L2 writing, several limitations must be noted to contextualize the findings and guide future research. One key limitation is the participant sample, which focused on intermediate to upper-intermediate learners in a specific academic setting. This may limit the generalizability of the findings to learners with different proficiency levels or in varied instructional contexts. Learners at beginning or advanced stages may interact differently with AI- or teacher-mediated feedback, warranting future research across a broader range of learner profiles. The feedback modalities used in this study, though reflective of typical instructional practice, also posed constraints. The AI tool lacked adaptive or interactive features, limiting its ability to emulate more dynamic, context-aware systems. Teacher-mediated feedback, while more

individualized, varies based on instructors' feedback styles, potentially influencing consistency. Future studies could benefit from more standardized feedback interventions and exploration of how different teacher feedback styles interact with learner characteristics.

Another limitation involves the genre of the writing task. This study focused exclusively on L2 narrative writing, which emphasizes creativity, fluency, and descriptive elaboration. Other genres—such as argumentative or analytical writing—may pose different demands and influence how learners respond to feedback. Including a wider range of genres in future research would provide a more comprehensive understanding of feedback effectiveness. The study's relatively short duration also limits conclusions about long-term learning. Although increases in accuracy and lexical sophistication were observed, the sustainability of these gains remains unclear. Longitudinal research incorporating delayed post-tests and follow-up writing tasks is needed to assess the lasting impact of AI- and teacher-mediated feedback.

A further limitation relates to the design of the AI feedback prompt. To maintain internal validity and comparability, the prompt deliberately omitted learner-specific context, such as proficiency levels, and did not assign a teaching role to the AI. While this standardized approach ensured consistency, it likely constrained the context-sensitivity of GPT-generated feedback compared to teacher feedback. Future research should investigate how integrating learner profiles and role-specific instructions into AI prompt design can enhance feedback personalization without compromising reliability or comparability.

Lastly, the study focused on product-based outcomes, without examining learners' engagement with feedback. Future research should incorporate qualitative methods, such as interviews, think-aloud protocols, or stimulated recall tasks to explore how learners respond to and apply different types of feedback. Understanding these cognitive and affective processes can deepen theoretical insights and inform practices that foster feedback literacy—learners' ability to evaluate and act upon the feedback they receive (Abdi Tabari & Johnson, 2025). Together, these avenues for future exploration can help refine hybrid feedback models, enhance learner engagement, and support sustainable improvements in L2 writing instruction.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Data availability statement

The data are confidential and cannot be shared.

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